

2026-08-19-anchor-block-integrity-176-findings-triage

2026-08-19 — Anchor-block-integrity checker surfaced 176 findings on the constitution mirror set: triage (no fixes)

Revision: 1 **Last modified:** 2026-08-19T11:49:41Z **Status:** active
Description: Triage of the 176 real findings BOB-105’s scripts/anchor_block_integrity_check.sh (commit 7cd080c on boba HEAD 0497b3d) surfaced against the four constitution mirrors (constitution/{CLAUDE, AGENTS, QWEN, GEMINI}.md). Investigation + classification ONLY per the operator directive — NO fixes applied. §11.4.238 discovery-channel-escape signal; §11.4.227(B) block-integrity/lockstep enforcement. **Authority:** §11.4.227(B) canonical vs. mirror layer discipline, §11.4.157 mirror lockstep, §11.4.238 automated-QA-discovers, §11.4.44 revision header, §11.4.6 no-guessing.

1. Summary counts (measured 2026-08-19T11:49:41Z)

Checker: bash scripts/anchor_block_integrity_check.sh --config scripts/anchor_block_integrity_check.conf (commit 7cd080c).

Category	Count
Total findings	176
DUPLICATE-OR-COLLISION	17
LOCKSTEP-DIVERGENCE	7 (task brief said 4 — actual 7)

Category	Count
LOCKSTEP-GAP	152

The 7 divergences (not 4): §11.4.69, §11.4.112, §11.4.120, §11.4.142, §11.4.192, §11.4.193, §11.4.201, §11.4.227. Re-counting the checker output: §11.4.201 is enumerated in the divergence list too. Correction: **8** distinct anchors show LOCKSTEP-DIVERGENCE when the additional duplicate-driven divergences are counted (see §3 note).

The 17 duplicates split: Constitution.md 3 (§11.4.30, §11.4.93, §11.4.202), CLAUDE.md 3, AGENTS.md 5, QWEN.md 4, GEMINI.md 2 — all in the anchor set {§11.4.69, .112, .120, .142, .201} plus the three Constitution-only cases.

2. DUPLICATE-OR-COLLISION — classification

All 17 duplicates are the SAME structural class: an anchor with (a) a compact-summary/reference-heading block **plus** (b) a later verbatim-mandate or amendment block. The F7 pattern the round-of-2026-07-22 amendment already forensically identified (§11.4.208/§11.4.209) is now measured on 8 additional anchors.

- **REAL DEFECT (mirror layer, 14 findings):**

§11.4.69/.112/.120/.142/.201 duplicated across CLAUDE/AGENTS/QWEN/GEMINI mirrors. First occurrence is the compact section (bulleted-list entry or short heading), second is the full amendment block from a later round. Per §11.4.227(B) exactly-once-per-file: >1 block-start = FAIL. Reconciliation follows the F7 pattern — keep the fuller variant, union in any unique element from the compact.

- **CANONICAL-VS-MIRROR LAYER (3 findings):** Constitution.md’s §11.4.30, §11.4.93, §11.4.202 each appear as two block-starts (### heading + a later **bold**). §11.4.227(B) explicitly permits canonical-vs-mirror layer variance; however the RULE “exactly one block-start per anchor per file” is written FILE-scoped, not layer-scoped — so these are REAL DEFECTS at the same severity as the mirror duplicates but arguably require operator adjudication whether Constitution.md’s dual-format tradition counts as a legitimate <H3>+bold-reference pair.

3. LOCKSTEP-DIVERGENCE — per-anchor triage

Extractor produces the first-block content-hash per mirror. Any mirror the same anchor appears twice in gets its FIRST occurrence hashed; that first occurrence's variant (compact vs. full) is often what the divergence reveals — not that the full text is different, but that the FIRST occurrence's shape differs.

3.1 §11.4.69 — REAL DEFECT

Hashes: AGENTS unique, others match. AGENTS.md has TWO occurrences (compact reference-heading L1526 + full amendment L2554); other three carry one block. First-occurrence hash for AGENTS is the compact short heading `**§11.4.69 – Universal sink-side positive-evidence taxonomy + mechanical enforcement (User mandate, 2026-05-20)**`; other mirrors' first hit is the full-body block. Same F7 shape as §11.4.208/.209.

3.2 §11.4.112 — REAL DEFECT

Three distinct hashes across four mirrors. AGENTS=QWEN duplicate the anchor (compact + full pattern), CLAUDE duplicates it too but at a different pair of line offsets, GEMINI is the outlier single-block. Root: divergent compact vs. verbatim variant of the impossibility-verdict-bounded amendment.

3.3 §11.4.120 — REAL DEFECT (same shape as .112)

Three hashes; AGENTS=QWEN, CLAUDE unique, GEMINI unique — compact/full order differs across the four mirrors.

3.4 §11.4.142 — REAL DEFECT

Two hashes; AGENTS=GEMINI=QWEN identical, CLAUDE differs. CLAUDE carries the “universal code-review — no exception” anchor as ONE block; the other three carry a second amendment block later in the file which the checker's first-block selection reveals as a compact/full mismatch.

3.5 §11.4.192 — REAL DEFECT (whitespace/trailer-only class)

Two hashes; AGENTS=CLAUDE, GEMINI=QWEN. Sampled bodies are word-for-word identical across 3200+ chars. Divergence is likely a trailing whitespace, trailing newline, or single-character formatting artifact (verified textually indistinguishable at 300-char boundaries). Fix is a byte-normalization pass; not a content re-mint.

3.6 §11.4.193 — REAL DEFECT (genuine content divergence)

Three hashes; QWEN=GEMINI, CLAUDE unique, AGENTS unique — the ONLY divergence of the seven that is a genuine content-level lockstep violation: * GEMINI+QWEN carry the **full verbatim mandate** (~2600 chars including operator quote + numbered clauses 1-6 + gate list). * CLAUDE.md carries a **medium-summary** variant (~700 chars, prose paragraph, no numbered clauses, no gate list). * AGENTS.md carries a **short-summary** variant (~350 chars: “Blind typing is STRICTLY FORBIDDEN everywhere — every agent, every tool, every automation...”). Three distinct information levels — the exact F7-class divergence §11.4.227 was minted to prevent, hitting an anchor minted AFTER §11.4.227 already existed. Needs operator reconciliation to the FULLEST variant (GEMINI/QWEN body).

3.7 §11.4.201 — REAL DEFECT

Duplicate + divergence in all four mirrors — a second amendment block landed in each mirror at a different line offset, and the checker’s first-hash selection picks up divergent variants.

3.8 §11.4.227 — REAL DEFECT (whitespace/trailer-only class)

Two hashes; AGENTS=GEMINI=QWEN identical, CLAUDE differs. Sampled bodies are near-identical (>5000 chars). Same likely root as §11.4.192 — trailing character or single-punctuation edit.

4. LOCKSTEP-GAP — 10-sample classification

Sampled across the 152 gap range. Categories per operator taxonomy:

Sample anchor	Present in	Classification	Notes
§11.4.1	CLAUDE only	REAL DEFECT	Anchor exists as ### §11.4.1 heading in CLAUDE but as un-shaped compact citation in others
§11.4.10.A	CLAUDE only	LEGITIMATE (sub-anchor)	Dotted sub-anchor legitimately only cataloged once
§11.4.83	AGENTS/ CLAUDE/ QWEN, NOT GEMINI	REAL DEFECT	Both ### (CLAUDE) and ** (AGENTS) block-start forms present; GEMINI genuinely lacks it
§11.4.100	AGENTS/QWEN only	CHECKER OVER-REACH + REAL DEFECT	CLAUDE has - §11.4.100 – RETIRED dash-bullet (regex excludes) — legitimate compact form; but GEMINI absent entirely
§11.4.108	AGENTS/ CLAUDE/ QWEN, NOT GEMINI	REAL DEFECT	GEMINI backfill needed — this is a core four-layer verification anchor
§11.4.135	AGENTS/QWEN only	REAL DEFECT	Missing from CLAUDE and GEMINI — release-

Sample anchor	Present in	Classification	Notes
§11.4.141	AGENTS/ GEMINI/QWEN, NOT CLAUDE	REAL DEFECT	blocking regression- guard anchor CLAUDE carries compact bullet (- §11.4.141 – ...) which regex excludes; historically known collision territory
§11.4.157	AGENTS/ GEMINI/QWEN, NOT CLAUDE	REAL DEFECT	The mirror- lockstep anchor itself is missing from one mirror — self-referential ledger drift
§11.4.192	(in gaps list)	CHECKER OVER-REACH	Also in divergence list — one finding double- reported
§11.4.72	AGENTS only	REAL DEFECT	Audio-top- priority anchor missing from 3 of 4 mirrors

Pattern extrapolation to the full 152

Three visible cohorts: 1. **§11.4.1..§11.4.99** — early-anchor era: CLAUDE.md is the “compact retelling” file (bulleted - §11.4.X entries), AGENTS.md is the “full mirror”. Regex `^(#{2,4} §|**§)` excludes dash-bullets → false-positive gap for anchors that ARE present in compact form. Estimated ~40-60 of the 152 are this class (checker over-reach — legitimate compact variance per §11.4.227(B) canonical-vs-mirror layer permission, extended to CLAUDE-vs-mirror layer by convention). 2. **§11.4.100..§11.4.134** — “present in AGENTS+CLAUDE+QWEN, NOT

GEMINI” cohort (~35 anchors). GEMINI.md was added LATER (§11.4.157 landed 2026-06-15) — anchors minted before GEMINI joined the mirror set were never backfilled. REAL DEFECT class per §11.4.157 lockstep, requires GEMINI backfill. 3. **§11.4.135..§11.4.165** — “present in AGENTS+GEMINI+QWEN, NOT CLAUDE” cohort (~30 anchors). CLAUDE.md’s compact-summary discipline for these ranges emits dash-bulleted - §11.4.X entries, not block-start-shaped headings. Legitimate compact variance OR checker over-reach depending on operator interpretation of §11.4.227(B).

Coarse split of the 152 gaps: ~70 **REAL DEFECT** (missing-from-GEMINI cohort + genuinely absent anchors), ~70 **CHECKER OVER-REACH / LEGITIMATE COMPACT VARIANCE** (CLAUDE dash-bullets), ~12 **sub-anchor edge cases** (§11.4.10.A-class dotted ids only cataloged once by design).

5. Recommended next-step operator decisions (NOT executed)

1. **Reconcile §11.4.193 (highest-severity)** — the only “3 information levels” divergence. Operator picks canonical variant (recommend GEMINI/QWEN full body) and lockstep-syncs to all 4 mirrors.
2. **Byte-normalize §11.4.192 and §11.4.227** — trailing-whitespace/newline pass; verifiable by re-running the checker until hashes converge.
3. **Reconcile the 5 duplicate-driven divergences** (§11.4.69/.112/.120/.142/.201) via the F7 recipe from the 2026-07-22 round: keep the fuller variant, union unique elements from the compact, delete the duplicate — per §11.4.227(B).
4. **Backfill GEMINI.md** with the §11.4.100..§11.4.134 range (~35 anchors) that predate its mirror inclusion — a bulk lockstep sync.
5. **Operator-decide the compact-vs-full policy question:** per §11.4.227(B) canonical-vs-mirror variance is legitimate; extend explicitly to CLAUDE-compact-vs-AGENTS-full, or require CLAUDE to carry block-start-shaped compacts (bold ****§11.4.X**** instead of dash - §11.4.X). Choice determines whether ~70 of 152 gaps are REAL DEFECT or checker over-reach.
6. **Checker tuning** — teach the extractor to (a) count dash-bulleted - §11.4.X as compact-form block-start (per operator’s policy in decision 5), (b) collapse dup-reports where an anchor appears in both DIVERGENCE and GAP lists, (c) surface Constitution.md dual **###+**** legitimately per §11.4.227(B).

7. **Fix Constitution.md's 3 duplicates** (§11.4.30/.93/.202) via the same F7 recipe OR document the dual-format as canonical-layer-permitted per §11.4.227(B).

None of the above are executed by this triage per operator directive.

6. Provenance

- Checker commit: 7cd080c (scripts/anchor_block_integrity_check.sh — BOB-105 landing)
- Repo HEAD at triage: 0497b3d
- Findings log preserved: /tmp/anchor_findings.log (186 lines, includes 176 FAIL lines + 10 header/verdict lines)
- Triage authored by: Claude (fresh agent, Opus 4.7)
- Discovery-channel-escape signal: §11.4.238 — the automated checker DID surface these; manual QA would have caught none of the whitespace-only or byte-hash divergences (§11.4.238 automated-QA-discovers holds true here).